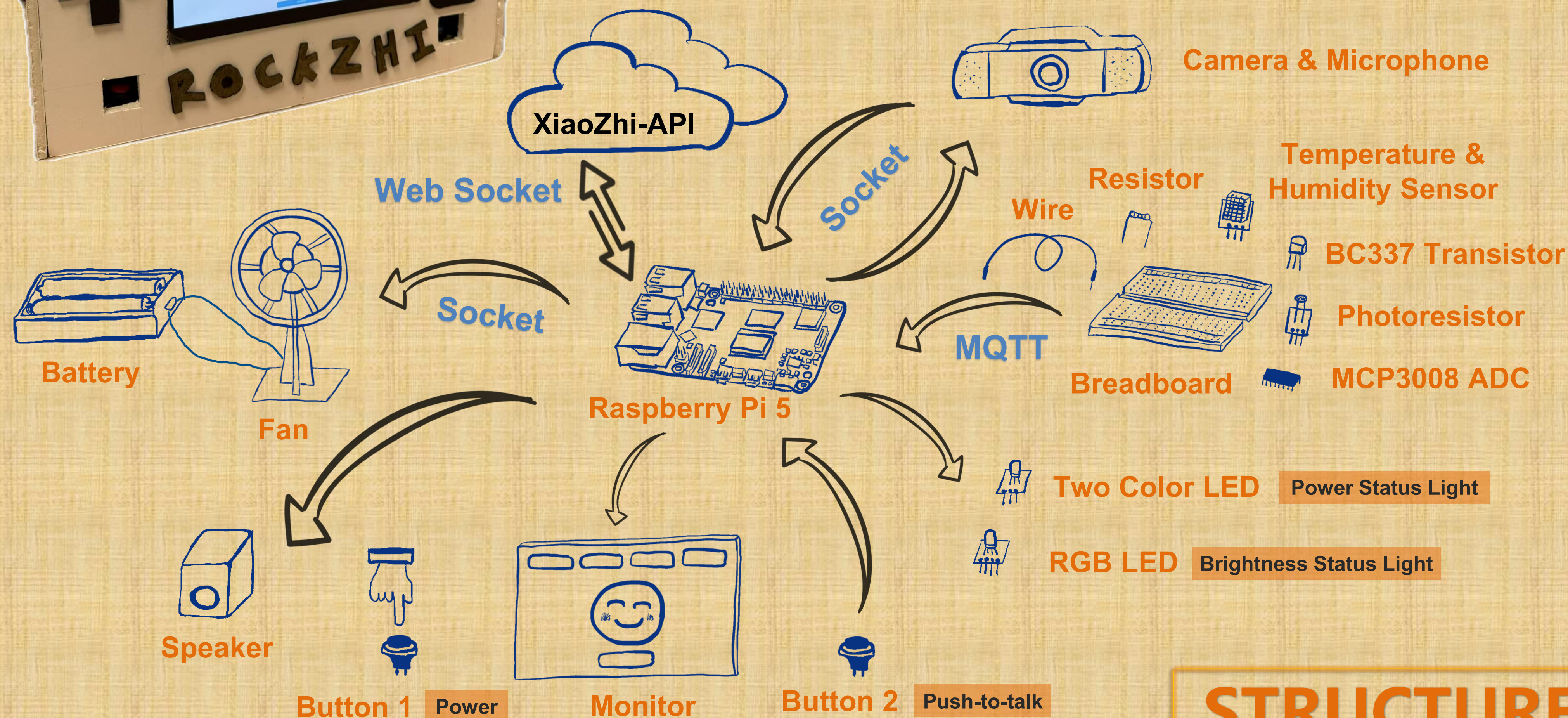




ROCKZHI

Talk, Play n' Health: The All-in-One Table Bot



STRUCTURE

PROBLEM

Children need companionship and smart learning support. Yet today's solutions are fragmented — smart speakers, apps, and health tools don't offer an integrated experience for emotional support, entertainment, and health during long desk hours.

SOLUTION

An all-in-one desktop robot powered by Raspberry Pi, equipped with a camera, microphone, and display for rich multimodal interaction. Featuring integrated AI and a smart voice assistant, it offers seamless emotional dialogue, interactive games, posture correction, and environmental sensing—creating a unified, intelligent companion that enhances both companionship and well-being during desk work.

TECHNICAL DESIGN

Artificial Intelligence

- **Gesture Model using MediaPipe Hands:** Recognizes hand signs like rock, paper, and scissors for interactive gameplay.
- **Posture Model using MediaPipe Pose:** Detects poor posture such as slouching by analyzing body angles.

Internet of Things

- **Environmental Sensors:** Measures temperature, humidity, and light in real time.
- **Smart Control:** Controls external devices like a fan.
- **Physical Buttons:** Offers quick-access buttons for power and voice activation.
- **Voice-Controlled:** All features accessible via ROCKZHI.